

Algebra Rules & Basic Calculus

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1. Factoring Trinomials (The “Multiply & Add” Game)

You saw the “Difference of Squares” (like $x^2 - 9$), which only has two pieces. But often, you will see expressions with *three* pieces, called a trinomial.

- **The Rule:** To factor something that looks like $x^2 + bx + c$, you need to find two numbers that **multiply** to make the last number (c) AND **add** up to make the middle number (b).
- **Example:** Factor $x^2 + 5x + 6$
 - *Question:* What multiplies to 6 and adds to 5?
 - *Answer:* 2 and 3. ($2 \times 3 = 6$, and $2 + 3 = 5$).
 - *Result:* $(x + 2)(x + 3)$
- **Why it’s needed:** Just like Difference of Squares, this is used constantly in Module 2 to cancel out “zero” denominators in limits.

2. Greatest Common Factor (GCF)

This is the simplest form of factoring, but the easiest to forget. It means pulling out the biggest piece that every term shares.

- **The Rule:** Look at all the pieces. What is the biggest number or letter they can all be divided by? Pull it out to the front.
- **Example:** $3x^2 - 12x$
 - *Question:* What do $3x^2$ and $12x$ have in common? Both can be divided by 3, and both have at least one x .
 - *Result:* Pull $3x$ to the front. You get $3x(x - 4)$.
- **Why it’s needed:** Often, doing this one simple step is enough to cross out a denominator and solve a limit.

3. The Exponent Rules (The Big Three)

When dealing with variables raised to powers, you need to know how those powers interact.

- **Rule A: Multiplying (Add the powers)**
 - If you multiply the same base, you add the tiny numbers.
 - *Example:* $x^2 \cdot x^3 = x^5$ (Because xx times xxx is $xxxxx$)
- **Rule B: Dividing (Subtract the powers)**
 - If you divide, the top power minus the bottom power wins.
 - *Example:*

$$\frac{x^5}{x^2} = x^3$$

- **Rule C: Negative Exponents (The Flip)**
 - A negative exponent is just a fraction in disguise. It means “move me to the bottom.”
 - *Example:* x^{-2} is exactly the same as

$$\frac{1}{x^2}$$

- **Why it’s needed:** When we get to Module 4 (Derivative Rules), knowing how to move exponents around is the entire key to solving the problems quickly!

4. Fractional Exponents (Hidden Roots)

- **The Rule:** A square root is actually just an exponent of $\frac{1}{2}$. A cube root is an exponent of $\frac{1}{3}$.
- **Example:** $\sqrt{x}$ is the exact same thing as $x^{1/2}$.
- **Why it's needed:** In calculus, there are no "rules" for square roots. So, whenever we see a square root, we rewrite it as a fraction exponent first, and then apply normal calculus rules.

5. The Conjugate (The Square Root Killer)

If you get a $\frac{0}{0}$ limit, but it has a square root in it, factoring won't work. You have to use a trick called the Conjugate.

- **The Rule:** The conjugate just means "flip the sign in the middle." If you have $(\sqrt{x} - 3)$, the conjugate is $(\sqrt{x} + 3)$. When you multiply them together, the square roots magically disappear.
 - **Example:** Multiply $(\sqrt{x} - 3)$ by its conjugate $(\sqrt{x} + 3)$.
 - *Steps:* Use FOIL (First, Outer, Inner, Last).
 - First: $\sqrt{x} \cdot \sqrt{x} = x$
 - Outer: $3\sqrt{x}$
 - Inner: $-3\sqrt{x}$ (Notice how the outer and inner cancel each other out to zero!)
 - Last: $-3 \cdot 3 = -9$
 - *Result:* $x - 9$
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